

HARSHITH NARASIMHAMURTHY

Data Scientist | ML Engineer | Generative AI

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PROFESSIONAL SUMMARY

Data Scientist with 3.5+ years building production ML systems across fintech, insurance, and agribusiness. Delivered 5 end-to-end ML products from geospatial credit scoring (~90% accuracy) to multi-agent GenAI systems with RAG and fine-tuned LLMs. Proven across the full lifecycle: feature engineering, model deployment (FastAPI/Docker), MLOps (Airflow/MLflow), and CI/CD on AWS.

PROFESSIONAL EXPERIENCE

Data Analyst 2 | [SatSure Analytics](#), Bengaluru, India

Jul 2023 – Present

- Built geospatial credit scoring model (Random Forest + satellite NDVI features) achieving ~90% accuracy in farmer creditworthiness; adopted by partner banks to accelerate agricultural loan underwriting.
- Designed insurance recovery pipeline integrating predictive risk scores with field-agent workflows, driving ~20% uplift in on-time loan collections (FY 2025–26).
- Automated ETL/ELT pipelines on Apache Airflow + AWS (S3, Athena, EC2), reducing data processing latency by ~25% and improving downstream analytics SLA reliability.
- Recognized as Customer Champion for sustained enterprise impact across 3+ banking and insurance clients through reliable model delivery and stakeholder collaboration.

ML Research Intern | [Western Sydney University](#), Sydney, Australia

Jan 2023 – Mar 2023

- Developed LSTM & ARIMA time-series forecasting models for crop yield prediction; engineered preprocessing and cross-validation workflows improving forecasting accuracy by ~12% over baseline.

KEY ML & GENAI PROJECTS

[NovaPay — Multi-Agent AI Customer Support System](#) | *Gemma-3 4B, QLoRA, RAG, LangChain, HuggingFace Spaces*

- Architected multi-agent orchestration with QLoRA fine-tuned triage routing to RAG-grounded specialists; scored 4.9/5.0 on LLM-as-Judge and handled 15/15 adversarial attacks, deployed live on HF Spaces at \$0 infra cost.
- Fine-tuned Gemma-3 4B (0.76% trainable params, 15 min on free T4) on 884 synthetic examples with quality-controlled data generation; built hallucination detection and provider-agnostic LLM client supporting 4 backends.
- Designed end-to-end evaluation: A/B/C comparison, BERTScore (F1 = 0.74), adversarial red-teaming, latency profiling, and 7-KPI ops dashboard; rigorously validated RAG's 10.6% faithfulness improvement over baseline.

[Formula 1 AI SQL Assistant — LLM-Powered Text-to-SQL Engine](#) | *Groq LLaMA 3.3 70B, LangChain, FastAPI*

- Built production Text-to-SQL agent with conversation memory, self-correction logic, and SELECT-only guardrails; enables zero-SQL natural language querying over live structured F1 race data via REST API.

[Customer Segmentation MLOps Pipeline — End-to-End ML System](#) | *KMeans, MLflow, DagsHub, FastAPI, Streamlit, Docker*

- Architected full MLOps lifecycle for RFM + behavioral customer segmentation; tracked experiments via MLflow + DagsHub for reproducible model comparison across runs.
- Deployed via FastAPI + Streamlit, containerized with Docker, automated CI/CD to Docker Hub via GitHub Actions; modular ingestion → validation → transformation → training pipeline from MongoDB with pytest coverage. Live on Streamlit Cloud.

[Real-Time Crop Health Monitoring — Production Satellite ML System](#) | *NDVI, K-Means, Geospatial ML, AWS*

- Deployed 10m × 10m NDVI time-series platform with K-Means health classification (Good/Average/Poor); achieved ~95% damage-detection accuracy and adopted by banks and insurers for loan underwriting and yield assurance at scale.

TECHNICAL SKILLS

Languages: Python, SQL, Bash

ML / DL: Scikit-learn, PyTorch, Transformers, PEFT/QLoRA, LSTM, Random Forest, XGBoost, Clustering, A/B Testing

Generative AI: LLMs, RAG, LangChain, HuggingFace, Prompt Engineering, Agentic AI, Fine-tuning

MLOps & Cloud: MLflow, Apache Airflow, Docker, GitHub Actions CI/CD, AWS (S3, Athena, EC2, ECR), FastAPI, Streamlit

Data & BI: Pandas, NumPy, Apache Iceberg, MongoDB, ETL/ELT, Tableau, Matplotlib, Seaborn

EDUCATION & RECOGNITION

- M.Sc. Agricultural Statistics | Anand Agricultural University | 2021–2023 | Gold Medalist (University Rank #1).
- B.Sc. Agriculture | University of Agricultural Sciences, Bengaluru | 2017–2021 | All India Rank #6.
- Certifications: Generative AI with LangChain & HuggingFace • HackerRank SQL (Advanced).
- Publications: Memory-based neural network for cumin price forecasting • RNN architecture for banana price forecasting.